CHyLL v2: Continuous Latent Embeddings of Hybrid Systems

What the model learns, and what each loss term enforces

Kaito Iwasaki

Hybrid Dynamical Systems

A *hybrid* dynamical system mixes smooth motion with sudden jumps.

- A walking robot: the swing leg moves smoothly, then strikes the ground and the state jumps.
- A bouncing ball: free flight under gravity, then an instantaneous change of velocity at each impact.

The jump is a genuine discontinuity. A smooth model of the dynamics cannot be fitted directly across it.

Three Example Systems

System	State dim.	Behavior
Rimless wheel	2	stable rolling limit cycle
Bouncing ball	2	bounces, losing energy at each impact
Compass-gait biped	4	period-doubling route to chaos

They increase in difficulty. The compass-gait biped walks with period 1, then 2, 4, 8, and finally chaotically as the ground slope steepens.

The Idea

Replace the discontinuous system with a continuous one that is easy to model.

- An **encoder** maps each state into a latent space.
- A single **smooth vector field** advances the latent state in time.
- A **decoder** maps the latent state back.

The jump is absorbed into the smooth latent flow. This follows Teng, Liu, and Sreenath (ICML 2026).

The Mapping-Cylinder State Space

To make the jump continuous, the state is augmented with one extra coordinate $s \in [0, 1]$:

$$y = (x, s) \in X'.$$

- $s = 0$: ordinary motion of the original system.
- s sweeping $0 \to 1$: the jump, unrolled as a continuous path.

On X' the trajectory is continuous, so a smooth latent model can be fitted to it.

The Learned Maps

CHyLL v2 learns three maps on the augmented state $y = (x, s) \in X'$:

$$
\begin{aligned}
E_\theta &: X' \to \mathbb{R}^m && \text{encoder,} \\
V_\theta &: \mathbb{R}^m \to \mathbb{R}^m && \text{latent vector field,} \\
D_\theta &: \mathbb{R}^m \to X' && \text{decoder.}
\end{aligned}
$$

The latent dimension is $m = 7$ for the rimless wheel and bouncing ball, and $m = 11$ for the compass-gait biped.

Training: The Forward Pass

For a sampled trajectory slice $y_0, \ldots, y_{T-1}$:

$$
\begin{aligned}
z_i &= E_\theta(y_i), \\
\widehat{z}_i &= \Psi_{i\tau}(z_0), \qquad \dot{z} = V_\theta(z), \\
\widehat{y}_i &= D_\theta(\widehat{z}_i).
\end{aligned}
$$

Each state is encoded, the latent flow is integrated forward, and the integrated latent trajectory is decoded. The loss compares the encoded z_i, the integrated $\widehat{z}_i$, and the decoded $\widehat{y}_i$ against the data.

The Commutative Diagram

Write Φ_τ for one time-τ step of the system on X', and Ψ_τ for the time-τ flow of $\dot{z} = V_\theta(z)$.

$$\begin{tikzcd}
X' \arrow[r, "\Phi_\tau"] \arrow[d, "E_\theta"'] & X' \arrow[d, shift left=1.5, "E_\theta"] \\
\mathbb{R}^m \arrow[r, "\Psi_\tau"'] & \mathbb{R}^m \arrow[u, shift left=1.5, "D_\theta"]
\end{tikzcd}$$

A faithful embedding makes this diagram commute:

- L_z forces the square to commute: $\Psi_\tau(E_\theta y) = E_\theta(\Phi_\tau y)$.
- L_x forces the decoder to recover the state: $D_\theta(\widehat{z}_i) = y_i$.

The Mapping-Cylinder Seam

The diagram fixes the dynamics. Three more terms handle the hybrid structure, at the seam where s wraps from 1 back to 0:

- L_g: the two sides of a reset map to the *same* latent point, so the encoder respects the gluing.
- L_v: the latent velocity has no kink crossing a seam.
- L_c: the encoder keeps all m latent coordinates active and does not collapse them.

These seams are detected from the sampled coordinate s in the data, not from symbolic guard equations.

What Each Loss Term Enforces

Term	Enforces	Code helper
L_x	decoder recovers the state	`reconstruction_loss`
L_z	encode/flow square commutes	`latent_consistency_loss`
L_g	reset pairs share a latent point	`gluing_loss`
L_v	latent velocity continuous at seams	`seam_velocity_loss`
L_c	latent coordinates do not collapse	`anti_collapse_loss`

L_x and L_z are the two commutative-diagram conditions. L_g, L_v, and L_c are the extra conditions specific to the hybrid mapping-cylinder structure.

The Composite Objective

$$L = w_x L_x + w_z L_z + w_g L_g + w_v L_v + w_c L_c.$$

Setting	$(w_x, w_z, w_g, w_v, w_c)$
Rimless wheel, bouncing ball	$(1,\ 1,\ 3,\ 1,\ 1)$
Compass-gait biped	$(1,\ 1,\ 3,\ 0,\ 1)$

The exact values for any run are stored in its `config.json`.

Results

- **Rimless wheel.** The model reproduces the rolling limit cycle, and the learned embedding recovers its expected single topological cycle.
- **Compass-gait biped.** The cascade analysis recovers the period-doubling sequence exactly: the learned latent states form 2, 4, and 8 clusters at the period-2, -4, and -8 gaits, and disperse under chaos.
- **Bouncing ball.** The orbit spirals slowly toward rest rather than closing into a loop, so its topological signature is only partially recovered.

Where the Code Lives

Object	File
$E_\theta, V_\theta, D_\theta$	`networks.py`
latent ODE rollout	`ode.py`
loss terms L_x, L_z, L_g, L_v, L_c	`losses.py`
training loop	`train.py`
system simulators and reset maps	`systems/`

These files are the CHyLL v2 core library, under `chyll_v2/chyll_v2/`. Topology postprocessing lives under `chyll_v2/cycling_signature/`.

Summary

- A hybrid system is made continuous on the mapping cylinder X'.
- CHyLL v2 learns an encoder, a latent vector field, and a decoder.
- L_z and L_x make the encode/flow/decode diagram commute.
- L_g, L_v, and L_c enforce the hybrid seam structure.
- The learned embeddings recover the expected topology for the rimless wheel and the compass-gait cascade.